MyDesign Portfolio

Element A

Element A: Presentation and Justification of the Problem

Mr. O'Farrell and his students face the challenge of managing TV production lighting materials effectively. Currently, they use a light-reflecting board to ensure proper lighting and eliminate shadows during interviews around the school. However, their current method is not ideal. The board is either held by a student, which can be tiring and inconsistent, or clipped onto a stand—both options are awkward and hard to manage. There is a need for a new system that makes the process simpler by securely attaching the reflective board to a stand, allowing it to reflect light from different angles without requiring extra effort or people to hold it.

Mr. O'Farrell's primary concern is creating a design that securely attaches the whiteboard to a stand or light pole, ensuring stability and ease of adjustment. In the past, students have had to manually position the board, which results in poor light reflection and uneven results. The current setup uses two clips on a stand, but it's not efficient, and purchasing a dedicated lighting system can be expensive, often costing over $150. The goal is to design a solution that stays within a $50 budget, is easy to use, portable, and sturdy enough to hold a reflective whiteboard (24 x 36 x 1 inches) weighing up to 230 grams without shifting or becoming unstable.

While not a top priority, a "finger"-shaped holding system for the board would be a nice feature. Additionally, adding a logo with the school's mascot could help show school pride. This new system would not only benefit Mr. O'Farrell's TV production class but also the WHS news team, film festival participants, and film class. The goal is to create a practical, cost-effective, and efficient solution that improves the overall lighting setups for interviews and filming activities.

Mr. O'Farrell and the TV Production Crew lack a suitable mechanism to hold his light reflective board, forcing him to either rely on a student or use a cumbersome, self-developed solution, both of which are inefficient. The goal of this design project is to create a system that allows a white board to reflect light onto a specific area. The key function is to make sure the board can be easily adjusted to reflect light from different

angles. Another important feature is the ability to securely attach the board to a light pole, ensuring it stays in place. The design should be adjustable, sturdy, and compact so it's easy to store and move around. Additionally, the system needs to support a reflective white board weighing up to 230 grams and fits within 60cmx60cm15 cm without worrying about it becoming unstable or shifting. These goals and limitations will help guide the creation of a practical and effective solution.

Stakeholders:-

1: Peter O'Farrell - Walpole High School TV Production teacher; main stakeholder who asked for project

2: TV Production Students (Secondary) - students of Mr O'Farrell's TV Production class; need mics to record news

3: Walpole High School Students - Primary stakeholder getting interviewed

Figure 1: WHS Timber TV. "Ambush News: Birds." YouTube, 26 Mar. 2024, http://www.youtube.com/watch?v=HvcdipWhHeE	Figure 3: lamp on light while holding reflector board (@TV Studio. 1/24/25)

Figure 2: Clamp holding up reflection board (@TV Studio. 1/24/25

Figure 4: : Clamp holding onto light (@TV Studio WHS, 1/24/25)

Figure 5: This is what it looks like when board and light are disconnected(what we are not supposed to do)(@TV Studio WHS, 1/24/25)

This is a problem we can solve, and we hope it will help many people. The mechanism we're creating won't just assist Mr. O'Farrell, but every student in the TV Production class. Right now, students always have to carry the light and reflector board separately, which is uncomfortable and inconvenient. Our goal is to make their work easier by designing a durable, adjustable clamp that will combine the light and reflector board into one easy-to-carry system.